Ch 3.2: Multiple Linear Regression

Lecture 6 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Jan 26, 2026

Announcements

Last time:

- 3.1 (Simple) linear regression

Announcements:

- Homework #2 Due Sunday 2/8

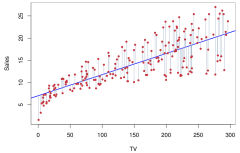
Covered in this lecture

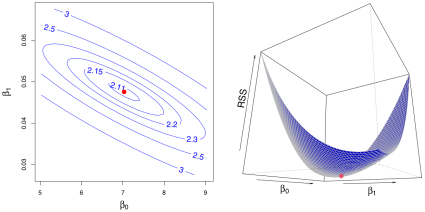
- Multiple linear regression
- Hypothesis test in that case
- Forward and Backward Selection

Section 1

Review from last time

Linear Regression with One Variable





- Predict Y on a single variable X

$$Y \approx \beta_0 + \beta_1 X$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
- $e_i = y_i - \hat{y}_i$ is the ith residual
- residual sum of squares RSS $= \sum_i e_i^2$

- RSS is minimized at *least squares coefficient estimates*

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Evaluating the model

- Linear regression is unbiased
- Variance of linear regression estimates:

$$\mathrm{SE}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Estimate σ: $\hat{\sigma}^2 = \frac{RSS}{n-2}$

- The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

- Hypothesis test:

  H_0: $\beta_1 = 0$
  H_a: $\beta_1 \neq 0$

  - Test statistic $t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$

Assessing the accuracy of the model

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$

an estimate of σ.
also measure lack of fit?
($MSE = RSS/n$)

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

$$TSS = \sum_i (y_i - \overline{y})^2$$

- TSS = total sum of squares
- Proportion in [0,1]; variability in Y that can be explained using X
- 0 means Y not explained by linear regression on X

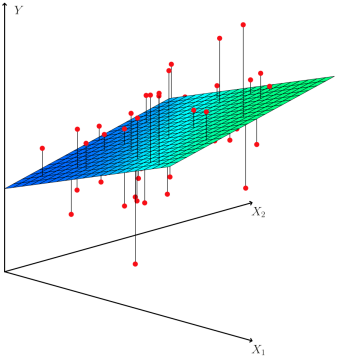
Section 2

Multiple Linear Regression

Setup

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p + \varepsilon$$

- β_j is avg affect on Y of one unit increase in X_j, holding all other predictors fixed.



Estimation and Prediction

Given estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \cdots, \hat{\beta}_p$, prediction is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Minimize the sum of squares

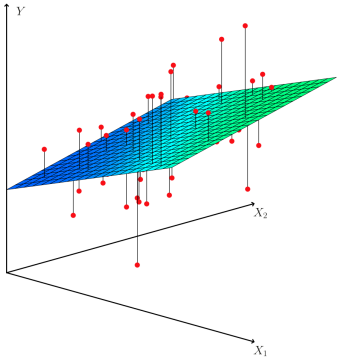
$$\begin{aligned} RSS &= \sum_i (y_i - \hat{y}_i)^2 \\ &= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip})^2 \end{aligned}$$

Coefficients are closed form but UGLY

We won't write them down, your favorite code can do this for us

Advertising data set example

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

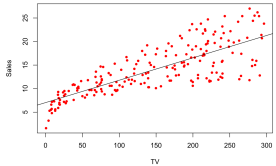
Interpretation of coefficients

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

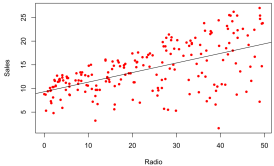
	Coefficient
`Intercept`	2.939
`TV`	0.046
`radio`	0.189
`newspaper`	−0.001

- Fixing TV and newspaper spending; Spending $1K more on radio results in 189 units additional sales
- What's going on with newspaper? This says no relationship between newspaper and sales

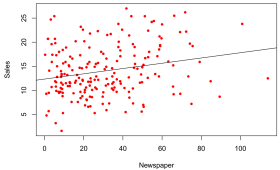
Single regression vs multi-regression



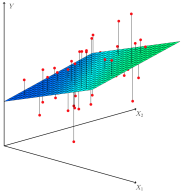
	Coefficient
Intercept	7.0325
TV	0.0475



	Coefficient
Intercept	9.312
radio	0.203



	Coefficient
Intercept	12.351
newspaper	0.055



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

Correlation matrix

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

- Correlation between radio and newspaper high (0.35)
- Markets with lots of radio also have lots of newspaper ads
- In single reg, newspaper gets credit for radio

Coding group work

Run the section titled "Multiple Linear Regression"

Section 3

Ch 3.2.2: Questions to ask of your regression

Question 1

Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response?

Q1: Hypothesis test

$H_0: \ \beta_1 = \beta_2 = \cdots = \beta_p = 0$

$H_a: \ \text{At least one } \beta_j \text{ is non-zero}$

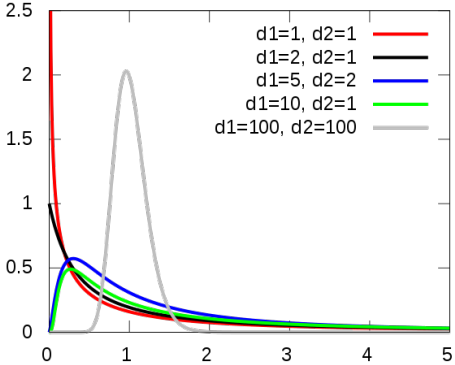
F-statistic:

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$

- $\text{TSS} = \sum_i (y_i - \overline{y})^2$
- $\text{RSS} = \sum_i (y_i - \hat{y}_i)^2$

The F-statistic for the hypothesis test

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$



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- If linear model assumptions are true

$$E(RSS/(n-p-1)) = \sigma^2$$

- and if H_0 is true

$$E((TSS - RSS)/p) = \sigma^2$$

- So if no relationship between response and predictors, F is close to 1
- If H_a is true,

$$E((TSS - RSS)/p) > \sigma^2$$

so then $F > 1$

Do Q1 section in jupyter notebook

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	2.939	0.3119	9.42	< 0.0001
`TV`	0.046	0.0014	32.81	< 0.0001
`radio`	0.189	0.0086	21.89	< 0.0001
`newspaper`	-0.001	0.0059	-0.18	0.8599

Quantity	Value
Residual standard error	1.69
R^2	0.897
F-statistic	570

- F stat is way bigger than 1, so reject null hypothesis
- How big is big enough? Use the p-value. Not included here but it's basically 0, so we're good

- Can also point out that how far from 1 required depends on the p, n values.
- If n large, an F-stat a little larger than 1 is still enough evidence against H_0
- However, if n small, might need to be further from 1 to be convinced.

Q2

Do all the predictors help to explain Y, or is only a subset of the predictors useful?

Q2: A first idea

Great, you know at least one variable is important, so which is it?....

- Bad idea plan A: All subsets or best subsets regression
  - Take every possible subset of p variables
  - Figure out least squares fit
  - Choose the best one based on some criterion balancing training error with model size

Do Q2 section in jupyter notebook

Why is this a bad idea?

- Why bad idea: $p = 40$ variables, this is checking more than a billion models
- Better idea: Automated appbroach that searches the subsets for a good one without checking all of them

- Later in the class, we will discuss better options for this (Ch 6, $\sim$ a month from now)
- Forward selection/ Backward selection to get subsets of variables without checking every possible subset
- Shrinkage methods (Lasso/Ridge)

Next time

CMSE381_S2026_Schedule : Sheet1

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression				
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	